

MATH 1075 - Section 7.6 Worksheet

09/24/2026

Solving a Rational Equation

Step 1 Factor the denominators of all rational expressions. Identify the restricted values.

Step 2 Identify the LCD of all expressions in the equation.

Step 3 Multiply both sides of the equation by the LCD.

Step 4 Solve the resulting equation.

Step 5 Check potential solutions in the original equation.

Solve following rational equations

1. $\frac{x}{2} + \frac{1}{3} = 2$

2. $\frac{2}{7} - \frac{1}{x} = \frac{2}{3}$

3. $\frac{2}{x+2} = \frac{5}{3x}$

4. $\frac{t}{12} + \frac{t+3}{3t} = \frac{1}{t}$

5. $\frac{x}{x+1} - 2 = \frac{-1}{x+1}$

$$6. \frac{4x}{x+3} - \frac{12}{x-3} = \frac{4x^2+36}{x^2-9}$$

$$7. \frac{6}{5a+10} - \frac{1}{a-5} = \frac{4}{a^2-3a-10}$$